

Goofy Symmetry and the Hierarchy Problem

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References

- [1] J. de Boer, F. Goertz, and A. Incrocci, “The goofy-symmetric Standard Model and the Hierarchy Problem,” *arXiv:2507.22111*.
- [2] A. Trautner, “Goofy transformations and the hierarchy problem,” *arXiv:2505.00099*.
- [3] L. Di Carlo et al., “Monte Carlo, blocking, and inference: How to measure the renormalization group flow,” *arXiv:2401.04811*.